

D205: Data Acquisition

Hannah R. Bergman

WGU Email: hbergm3@wgu.edu

Table of Contents

Part A: Research Question	2
A1: Question Data: Parameters, Definitions & Data Sources	2
Part B: Entity Relationship Diagram (ERD)	3
B1: SQL Code: Create New Table	4
B2: SQL Code: Load Data From CSV Into New Table	5
Part C: Query Results	5
C1: Data File: Results From the SQL Query	6
Part D: Time Period	7
D1: Why Monthly Updates Are Relevant To Our Business Needs	7
Part E: Demo Video Recording	7
E1: Panopto Video Recording Link:	7
Part F: Data Source	7
Part G: References	7

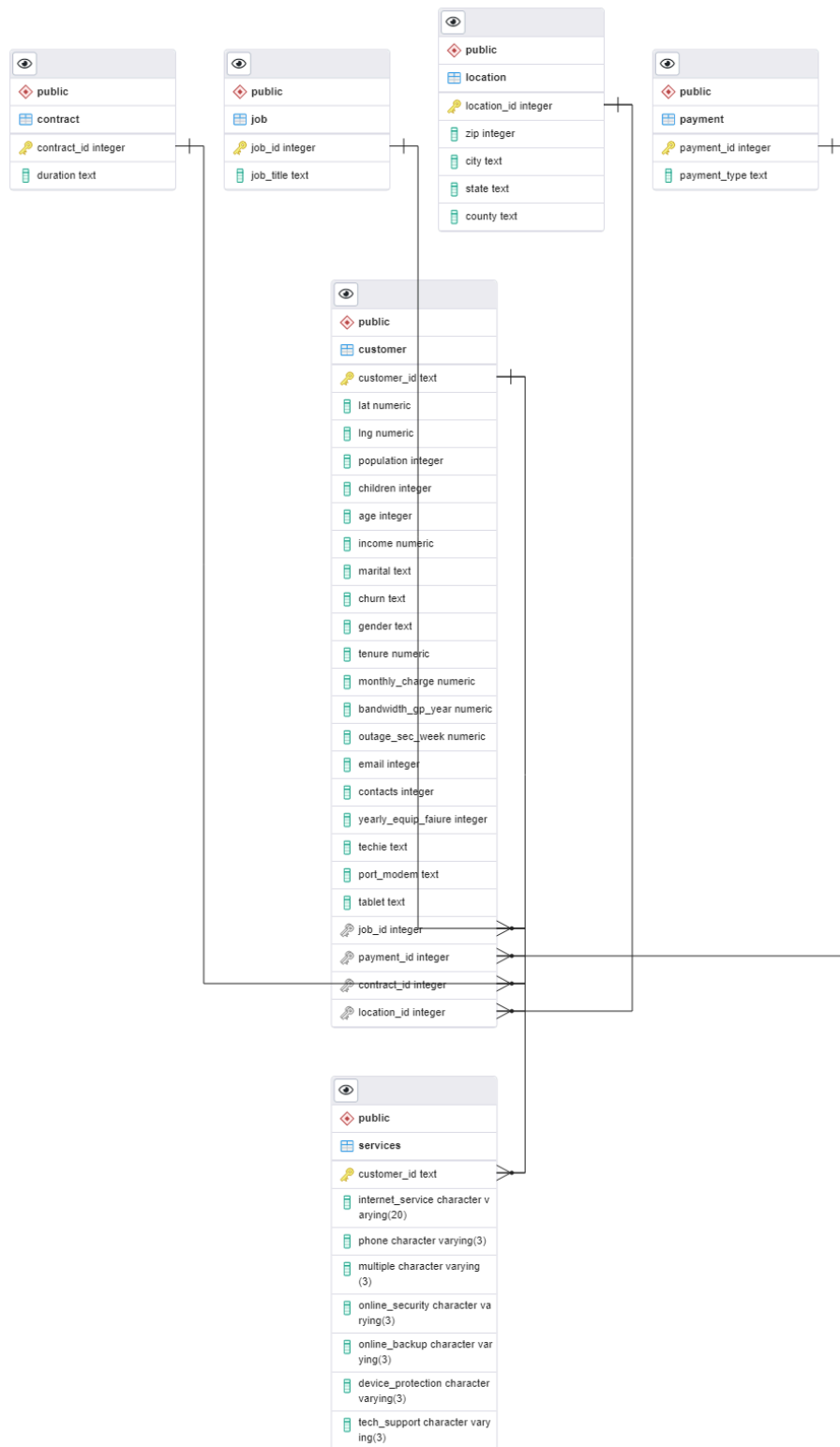
Part A: Research Question

Among **female customers aged 35 and under**, how many **use our Tech Support services**?

A1: Question Data: Parameters, Definitions & Data Sources

- **Female customers under 35:** In this study, we restrict the focus of our analysis to female participants who self-identify their gender as 'Female'. To facilitate audience segmentation and explore the specific usage patterns of younger demographics within our tech support user base, we further limit the scope to female participants aged 35 and under.
 - **Data Sourcing:**
 - Table:
 - customer (data from the original database)
 - Columns:
 - age (data type: integer)
 - gender (data type: text)
 - customer_id (Primary Key, data type: text)
- **Use our Tech Support services:** Using collected customer data, our query aims to analyze the adoption rate of Tech Support services among female customers aged 35 and under. By filtering for 'Yes' entries in the 'tech_support' field, we'll isolate the number of subscribed individuals within this demographic segment. We'll further compare this figure to the total number of female customers in the same age group, calculating the percentage of customers subscribing to Tech Support within this specific demographic segment.
 - **Data Sourcing:**
 - Table:
 - services (add-on CSV file data)
 - Columns:
 - tech_support (data type: varchar)
 - customer_id (Primary Key, data type: text)

Part B: Entity Relationship Diagram (ERD)



Relational Constraints

The services table in PostgreSQL was populated with data extracted from the services.csv file. To maintain data consistency and prevent errors, 'customer_id' was designated as the primary key, ensuring unique customer identification within the services table. To enforce relational integrity and prevent orphaned records, 'customer_id' was established as a foreign key, linking it to the existing customer table and requiring a customer's information to be first recorded in the customer table before associating services with them. This cross-table dependency is also the reason the text data type was selected for this column, aligning with its data type in the customer table and ensuring compatibility. Additionally, a NOT NULL constraint was applied to all fields, preventing the insertion of records with missing or invalid customer identifiers, thus upholding data integrity.

1:1 Relationship

The customer and services tables exhibit a one-to-one (1:1) relationship, with both tables containing 10,000 rows. This signifies that each customer record in the customer table is uniquely associated with a single set of service options stored in the services table, and vice versa, established through the customer_id field.

B1: SQL Code: Create New Table

```
CREATE TABLE public.services
(
  customer_id text NOT NULL,
  internet_service varchar(20) NOT NULL,
  phone varchar(3) NOT NULL,
  multiple varchar(3) NOT NULL,
  online_security varchar(3) NOT NULL,
  online_backup varchar(3) NOT NULL,
  device_protection varchar(3) NOT NULL,
  tech_support varchar(3) NOT NULL,
  PRIMARY KEY (customer_id),
  CONSTRAINT customer_id_fkey FOREIGN KEY (customer_id)
  REFERENCES public.customer (customer_id) MATCH SIMPLE
  ON UPDATE NO ACTION
  ON DELETE NO ACTION
  NOT VALID
)
TABLESPACE pg_default;

ALTER TABLE public.services
```

```
OWNER to postgres;
```

The process of database table creation and data loading involved manual execution of an SQL script within the PostgreSQL Query Tool (Query Editor). This script defines the structure of a new table called 'services' by declaring its columns, selecting data types for each column, and implementing primary key and foreign key constraints. Additionally, this code integrates data from an external services.csv file, populating the newly created table with relevant information.

B2: SQL Code: Load Data From CSV Into New Table

```
--command " "\\copy public.services (customer_id, internet_service, phone,
multiple, online_security, online_backup, device_protection, tech_support)
FROM 'C:/LabFiles/Services.csv' DELIMITER ',' CSV HEADER QUOTE '\"' ESCAPE
''';"
```

In order to append data from the services.csv file to the 'services' table, the PostgreSQL import/export functionality was employed, enabling a seamless transfer of information.

Part C: Query Results

```
-- SQL QUERY THAT ANSWERS THE RESEARCH QUESTION

SELECT
/* Select customer age */
    c.age AS customer_age,
/* Use the SUM function with a CASE statement to tally how many women under
the age of 35 have tech support */
    SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND s.tech_support =
'Yes' THEN 1 ELSE 0 END) AS female_customers_with_tech_support,
/* Use the COUNT function with a CASE statement to count the total number
of women under 35 with or without tech support */
    COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN c.customer_id
END) AS total_female_customers,
/* Use the COUNT and SUM functions together with a CASE statement to
calculate the percentage of female customers under 35 who have tech support
*/
    CASE WHEN COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN
c.customer_id END) > 0
    THEN (CAST(SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND
s.tech_support = 'Yes' THEN 1 ELSE 0 END) AS DECIMAL) / COUNT(CASE WHEN
c.gender = 'Female' AND c.age <= 35 THEN c.customer_id END)) * 100
```

```

        ELSE 0
    END AS percentage_female_customers_with_tech_support
FROM
    customer AS c
/* Use a Left Join to link the 2 tables, customer and services on
customer_id */
LEFT JOIN
    services AS s ON c.customer_id = s.customer_id
/* Filter customer records to extract those where the age is less than or
equal to 35 */
WHERE
    c.age <= 35
/* Group and Order By customer age in descending order, going from oldest
to youngest customers */
GROUP BY
    c.age
ORDER BY
    c.age DESC;

```

C1: Data File: Results From the SQL Query

The following screenshot provides a quick visual overview of the query's outcome:

Data Output		Explain	Messages	Notifications
	customer_age integer	female_customers_with_tech_support bigint	total_female_customers bigint	percentage_female_customers_with_tech_support numeric
1	35	22	79	27.84810126582278481000
2	34	21	67	31.34328358208955223900
3	33	21	60	35.00000000000000000000
4	32	30	77	38.96103896103896103900
5	31	25	71	35.21126760563380281700
6	30	21	66	31.81818181818181818200
7	29	28	83	33.73493975903614457800
8	28	21	72	29.166666666666666666700
9	27	27	69	39.13043478260869565200
10	26	35	75	46.666666666666666666700
11	25	21	66	31.81818181818181818200
12	24	32	80	40.00000000000000000000
13	23	25	68	36.76470588235294117600
14	22	22	60	36.666666666666666666700
15	21	29	71	40.84507042253521126800
16	20	32	72	44.444444444444444444400

For tabular exploration of the results, please refer to the CSV file included with the submission of this report, titled `women_with_tech_support.csv`.

Part D: Time Period

In alignment with our primary research question, our goal is to monitor how many of our female customers aged 35 and under utilize our Tech Support services. Due to our company's month-to-month contractual minimums, the database will require monthly updates with the latest service data to stay up-to-date.

D1: Why Monthly Updates Are Relevant To Our Business Needs

As far as future deadlines, our team suggests maintaining this ongoing research for at least 2 years, in correlation with our 2-year contractual maximums. This ongoing process will allow us to have a more comprehensive and accurate understanding of customer behavior, that we can utilize to perform actionable insights to improve customer acquisition and bolster customer loyalty with our company.

Part E: Demo Video Recording

E1: Panopto Video Recording Link:

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=e57f8f8f-5a74-45da-9360-b0f300360e2f>

Part F: Data Source

Western Governors University. (2023). *Churn Data and Dictionary Files* [Dataset]. Data Files and Associated Dictionary Files.

Retrieved Dec 11, 2023, from:

<https://access.wgu.edu/ASP3/aap/content/g9rke9s0rlc9ejd92md0.html>

Part G: References

Carchedi, N. (n.d.). *Intermediate SQL Queries* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/intermediate-sql-queries>

Gorenshteyn, D. (n.d.). *Applying SQL to Real-World Problems* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/applying-sql-to-real-world-problems>

Grossenbacher, T. (n.d.). *Introduction to Relational Databases in SQL* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/introduction-to-relational-databases-in-sql>

Ismay, C. (n.d.). *SQL for Joining Data* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/sql-for-joining-data>

Khalil, M. (n.d.). *Data Manipulation in SQL* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/data-manipulation-in-sql>

Malik, U., Goldwasser, M., & Johnston, B. (2019). *SQL for Data Analytics: Perform Fast and Efficient Data Analysis with the Power of SQL*. Packt Publishing, Limited.

Sewell, W. (2023, April). *SQL Sunday* [PowerPoint slides]. College of IT, Western Governors University.

Retrieved Dec 11, 2023, from:

https://westerngovernorsuniversity-my.sharepoint.com/:p:/g/personal/william_sewell_wgu_edu/EZvWh7S_6K9HnjS1yDoRXhQBBaL6oZZqrUM6TgsDBbbimQ?rttime=agStbWr620g

Sulmont, L. (n.d.). *Database Design* [Interactive Course]. Datacamp.

Retrieved Dec 11, 2023, from:

<https://app.datacamp.com/learn/courses/database-design>